

End-To-End Data Analytics Project Using SQL & Power BI

❖ Key Point Indicator (KPI's):

- **Total Sales:**

```
SELECT SUM(SalesAmount) AS "Revenue"
FROM [SalesData] JOIN [CustomerData] ON
[SalesData].[CustomerID] = [CustomerData].[CustomerID]
```

	Revenue
1	14164619

- **Total Orders:**

```
SELECT DISTINCT(COUNT(OrderID)) AS "Total Orders"
FROM [SalesData] JOIN [CustomerData] ON
[SalesData].[CustomerID] = [CustomerData].[CustomerID]
```

	Total Orders
1	186

- **Total Customers:**

```
SELECT DISTINCT(COUNT(CustomerID)) AS
"Total Customer" FROM
[SalesData]
```

	Total Customer
1	200

- **Gender Percentage:**

```
SELECT Gender, SUM(SalesAmount) AS "Revenue",
CONCAT(ROUND(SUM(SalesAmount)*100/SUM(SUM(SalesAmount)) OVER(),2), '%')
AS "Percentages" FROM [SalesData] JOIN [CustomerData] ON
[SalesData].[CustomerID]=[CustomerData].[CustomerID]

GROUP BY Gender
```

	Gender	Revenue	Percentages
1	Female	8641495	61.01%
2	Male	5523124	38.99%

- **Average Order Value (AOV) :**

```
SELECT ROUND((SUM(SalesAmount)*100/COUNT(OrderID)),0)
AS "AOV" FROM [SalesData] JOIN [CustomerData] ON
[SalesData].[CustomerID] = [CustomerData].[CustomerID]
```

	AOV
1	7615387

❖ Month-Wise Sales:

```
SELECT [Month Name],[Month],SUM(SalesAmount) AS "Revenue"
FROM SalesData JOIN CustomerData ON
SalesData.[CustomerID]=CustomerData.[CustomerID]
```

```
GROUP BY [Month Name],[Month]
ORDER BY [Month]
```

	Month Name	Month	Revenue
1	January	1	1043448
2	February	2	1612665
3	March	3	797125
4	April	4	1164472
5	May	5	1300993
6	June	6	984033
7	July	7	1611812
8	August	8	1138494
9	September	9	1648210
10	October	10	903924
11	November	11	728489
12	December	12	1230954

❖ Week-Wise Sales:

```
SELECT [Day Name],DATEPART(WEEKDAY,OrderDate) AS "Week Number",
SUM(SalesAmount) AS "Revenue"
FROM [SalesData] JOIN [CustomerData] ON
[SalesData].[CustomerID] =[CustomerData].[CustomerID]
```

```
GROUP BY [Day Name],DATEPART(WEEKDAY,OrderDate)
ORDER BY DATEPART(WEEKDAY,OrderDate)
```

	Day Name	Week Number	Revenue
1	Sunday	1	2336420
2	Monday	2	2145572
3	Tuesday	3	1785381
4	Wednesday	4	1971924
5	Thursday	5	1189327
6	Friday	6	2468894
7	Saturday	7	2267101

❖ Cities-Wise Sales :

```
SELECT City, SUM(SalesAmount) AS "Revenue"
FROM [SalesData] JOIN [CustomerData] ON
[SalesData].[CustomerID] =[CustomerData].[CustomerID]
GROUP BY City
ORDER BY Revenue DESC
```

	City	Revenue
1	Quetta	3656183
2	Islamabad	3303699
3	Lahore	2695612
4	Peshawar	2264110
5	Karachi	2245015

❖ Top 5 Customers:

```
SELECT Top 5 CustomerName, SUM(SalesAmount) AS "Revenue"
FROM [SalesData] JOIN [CustomerData] ON
[SalesData].[CustomerID] = [CustomerData].[CustomerID]

GROUP BY CustomerName
ORDER BY Revenue DESC
```

	CustomerName	Revenue
1	Customer_112	496305
2	Customer_66	453795
3	Customer_13	418167
4	Customer_30	373949
5	Customer_88	367843

❖ Payment Mode:

```
SELECT PaymentMode ,
CONCAT(ROUND(SUM(SalesAmount)*100/SUM(SUM(SalesAmount))) OVER(),2), '%')
AS "Percent" FROM [SalesData] JOIN [CustomerData] ON
[SalesData].[CustomerID] =[CustomerData].[CustomerID]

GROUP BY PaymentMode
```

	PaymentMode	Percent
1	Card	27.96%
2	Cash	37.48%
3	Online	34.56%

❖ Product - Wise Sales:

```
SELECT Product, SUM(SalesAmount) AS "Revenue"
FROM [SalesData] JOIN [CustomerData] ON
[SalesData].[CustomerID] =[CustomerData].[CustomerID]

GROUP BY Product
ORDER BY Revenue DESC
```

	Product	Revenue
1	Mouse	2728901
2	Printer	2414209
3	Monitor	2254366
4	Laptop	2021485
5	Tablet	1935090
6	Keyboard	1460562
7	Mobile	1350006

❖ Age-Distribution:

```
SELECT
CASE
WHEN Age>18 AND Age <=20 THEN 'Under 20'
WHEN Age>21 AND Age <=30 THEN 'Under 30'
WHEN Age>31 AND Age <=40 THEN 'Under 40'
WHEN Age>41 AND Age <=50 THEN 'Under 50'
ELSE
'Under60' END AS "Age Distribution",
COUNT(OrderID) AS "Orders" ,
SUM(SalesAmount) AS "Revenue"
FROM [SalesData] JOIN [CustomerData] ON
[SalesData].[CustomerID] = [CustomerData].[CustomerID]

GROUP BY
CASE
WHEN Age>18 AND Age <=20 THEN 'Under 20'
WHEN Age>21 AND Age <=30 THEN 'Under 30'
WHEN Age>31 AND Age <=40 THEN 'Under 40'
WHEN Age>41 AND Age <=50 THEN 'Under 50'
ELSE
'Under60' END
ORDER BY Orders DESC
```

	Age Distribution	Orders	Revenue
1	Under60	51	3682119
2	Under 50	47	3672642
3	Under 30	42	3463826
4	Under 40	40	2885728
5	Under 20	6	460304

❖ Week-Distribution :

```

SELECT
    DATEADD(DAY, 1 - DATEPART(WEEKDAY, OrderDate), CAST(OrderDate AS DATE))
AS WeekStart,
    DATEADD(DAY, 7 - DATEPART(WEEKDAY, OrderDate), CAST(OrderDate AS DATE))
AS WeekEnd,
    SUM(SalesAmount) AS "Revenue"
FROM [SalesData]
JOIN [CustomerData]
    ON [SalesData].[CustomerID] =[SalesData].[CustomerID]
GROUP BY
    DATEADD(DAY, 1 - DATEPART(WEEKDAY, OrderDate), CAST(OrderDate AS DATE)),
    DATEADD(DAY, 7 - DATEPART(WEEKDAY, OrderDate), CAST(OrderDate AS DATE))
ORDER BY WeekStart

```

	Week Start	Week End	Revenue
1	2023-01-08	2023-01-14	1778640
2	2023-01-15	2023-01-21	44179800
3	2023-01-22	2023-01-28	48163680
4	2023-01-29	2023-02-04	50414640
5	2023-02-05	2023-02-11	49416960
6	2023-02-12	2023-02-18	45347760
7	2023-02-19	2023-02-25	50192640
8	2023-02-26	2023-03-04	45094680
9	2023-03-05	2023-03-11	14919360
10	2023-03-12	2023-03-18	6993360
11	2023-03-19	2023-03-25	32218560
12	2023-03-26	2023-04-01	16987920
13	2023-04-02	2023-04-08	80624520
14	2023-04-09	2023-04-15	18314280
15	2023-04-16	2023-04-22	3037200
16	2023-04-23	2023-04-29	41784000
17	2023-04-30	2023-05-06	49427520
18	2023-05-07	2023-05-13	20925360
19	2023-05-14	2023-05-20	37704240

20	2023-05-21	2023-05-27	19366320
21	2023-05-28	2023-06-03	53112240
22	2023-06-04	2023-06-10	25128000
23	2023-06-11	2023-06-17	24848760
24	2023-06-18	2023-06-24	29593320
25	2023-06-25	2023-07-01	66480840
26	2023-07-02	2023-07-08	43005720
27	2023-07-09	2023-07-15	34836480
28	2023-07-16	2023-07-22	29789640
29	2023-07-23	2023-07-29	46348320
30	2023-07-30	2023-08-05	82681680
31	2023-08-06	2023-08-12	23618760
32	2023-08-13	2023-08-19	46606080
33	2023-08-20	2023-08-26	30228000
34	2023-08-27	2023-09-02	45858600
35	2023-09-03	2023-09-09	32300400
36	2023-09-10	2023-09-16	47259480
37	2023-09-17	2023-09-23	36522960
38	2023-09-24	2023-09-30	55782480

39	2023-10-01	2023-10-07	65943360
40	2023-10-08	2023-10-14	9668280
41	2023-10-15	2023-10-21	25524720
42	2023-10-22	2023-10-28	7334520
43	2023-10-29	2023-11-04	25462920
44	2023-11-05	2023-11-11	19686600
45	2023-11-12	2023-11-18	3321360
46	2023-11-19	2023-11-25	40704120
47	2023-11-26	2023-12-02	21323400
48	2023-12-03	2023-12-09	60774600
49	2023-12-10	2023-12-16	16610520
50	2023-12-17	2023-12-23	10934640
51	2023-12-24	2023-12-30	24051000
52	2023-12-31	2024-01-06	58664760

❖ Running Total:

```

SELECT [Month Name],
DATEPART(MONTH,OrderDate) AS "MonthNumber",SUM(SalesAmount) AS "Revenue",
SUM(SUM(SalesAmount)) OVER (ORDER BY DATEPART(MONTH,OrderDate)) AS "Running
Total"
FROM [SalesData] JOIN [CustomerData] ON [SalesData].[CustomerID]
=[CustomerData].[CustomerID]

GROUP BY [Month Name],
DATEPART(MONTH,OrderDate)
ORDER BY MonthNumber

```

	Month Name	MonthNumber	Revenue	Running Total
1	January	1	1043448	1043448
2	February	2	1612665	2656113
3	March	3	797125	3453238
4	April	4	1164472	4617710
5	May	5	1300993	5918703
6	June	6	984033	6902736
7	July	7	1611812	8514548
8	August	8	1138494	9653042
9	September	9	1648210	11301252
10	October	10	903924	12205176
11	November	11	728489	12933665
12	December	12	1230954	14164619

❖ Month -Camparision:

```

SELECT [Month Name],
DATEPART(MONTH,OrderDate) AS "Month Number",
SUM(SalesAmount) AS "Revenue",
SUM(SalesAmount)-LAG(SUM(SalesAmount)) OVER(ORDER BY
DATEPART(MONTH,OrderDate))
AS "Month Diff",
CONCAT(ROUND((SUM(SalesAmount)-LAG(SUM(SalesAmount))
OVER(ORDER BY DATEPART(MONTH,OrderDate)))/
LAG(SUM(SalesAmount)) OVER(ORDER BY DATEPART(MONTH,OrderDate))*100,2), '%')
AS "Percentage"
FROM [SalesData] JOIN [CustomerData] ON
[SalesData].[CustomerID]=[CustomerData].[CustomerID]

GROUP BY [Month Name] ,
DATEPART(MONTH,OrderDate)

ORDER BY [Month Number]

```

	Month Name	Month Number	Revenue	Month Diff	Percentage
1	January	1	1043448	NULL	%
2	February	2	1612665	569217	54.55%
3	March	3	797125	-815540	-50.57%
4	April	4	1164472	367347	46.08%
5	May	5	1300993	136521	11.72%
6	June	6	984033	-316960	-24.36%
7	July	7	1611812	627779	63.8%
8	August	8	1138494	-473318	-29.37%
9	September	9	1648210	509716	44.77%
10	October	10	903924	-744286	-45.16%
11	November	11	728489	-175435	-19.41%
12	December	12	1230954	502465	68.97%

